

Changing the `veg` also change `a`.

- Key Point:
- ① must be initialized at declaration.
 - ② cannot be change to refer another variable.
 - ③ work like same variable (no new-memory)

Typocasting

Typocasting means converting one datatype to another.

① implicit Type casting (Automatic)

```
int a = 5;
```

```
double b = a; // int → double.
```

done by compiler automatically

② Explicit Type casting (Manual)

(a) C style `double x = 5.7`
`int y = (int)x;`

(b) C++ style `double x = 5.7`
`int y = static_cast<int>(x);`

③ dynamic cast for Polymorphism.

```
base * b = new Derived();
```

```
Derived * d = dynamic_cast<Derived*>(b);
```

use with virtual function.

④ Const-Cast

```
const int a = 100;
```

```
int * ptr = const_cast<int*>(a);
```

Remove Const.

Problem statement 1: Employee salary system

- Create program base class Employee with virtual salary()
- Derive full time and Intern and Override it
- Use Base pointer to call salary

Problem statement 2: Shape Area.

- Create base class shape with virtual area()
- Derive Circle and Rectangle and Override it
- Use Base pointer to find covered area

Problem statement 3: Animal sound.

- Create base class Animal with virtual sound()
- Derive Dog and Cat and override it
- Store object in base pointer array and call sound() using loop.

Day 44 C Reference variable and Type casting

A Reference variable is an alias (another name) for an existing variable.

Syntax:

```
int a = 10;
```

```
int &ref = a;
```

Example:

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a = 5;
```

```
    int &ref = a;
```

```
    ref = 10;
```

```
    cout << a; // 10
```


⑧ Reinterpret_Cast

```
int a = 65;  
char *p = reinterpret_cast<char*>(&a);  
#Low level Conversation.
```

Problem Statement 1: Reference variable basic.

Problem Statement 2: swap using Reference

Problem Statement 3: type casting in polymorphism

Create: • A base class shape with virtual function
Derived class Circle:

Task: • Create base pointer pointing to user.
• Using dynamic cast to convert it
int a derived pointer
• create derived pointer.